

The University of Texas at Austin

HOMEWORK ASSIGNMENT 7

M339D Introduction to Financial Mathematics

October 10, 2026

Instructions: Provide your complete solution to the following problems. Final answers only, without appropriate justification, will receive zero points even if correct.

PUT-CALL PARITY.

Problem 7.1. (5 points) A certain common stock is priced at \$42.00 per share. Assume that the continuously compounded interest rate is $r = 10.00\%$ per annum. Consider a \$50 –strike European call, maturing in 3 years which currently sells for \$10.80. What is the price of the corresponding 3 –year, \$50 –strike European put option?

- a. \$5.20
- b. \$5.69
- c. \$5.04
- d. \$5.84
- e. None of the above.

Problem 7.2. (5 points) The initial price of a non-dividend-paying stock is \$55 per share. A 6 –month, at-the-money call option is trading for \$1.89. Let the interest rate be $r = 0.065$. Find the price of the European put with the same strike, expiration and the underlying asset.

- a. \$0.05
- b. \$0.13
- c. \$0.56
- d. \$0.88
- e. None of the above

Problem 7.3. (5 points) *Source: Problem #2 from the Sample IFM (Derivatives: Introductory) questions.*

You are given the following information:

- The current price to buy one share of XYZ stock is \$500.
- The stock does not pay dividends.
- The risk-free interest rate, compounded continuously, is 6%.
- A European call option on one share of XYZ stock with a strike price of K that expires in one year costs \$66.59.
- A European put option on one share of XYZ stock with a strike price of K that expires in one year costs \$18.64.

Determine the strike price K .

- a. \$449
- b. \$452
- c. \$480
- d. \$559
- e. None of the above.

Problem 7.4. (5 points) Consider a European call option and a European put option on a non-dividend-paying stock. Assume:

- The current price of the stock is \$55.
- The call option currently sells for \$0.15 more than the put option.
- Both options expire in 4 years.
- Both options have a strike price of \$70.

Calculate the continuously compounded risk-free interest rate r .

- a. 0.044
- b. 0.052
- c. 0.06
- d. 0.065
- e. None of the above.

Problem 7.5. (5 points) Consider a European call option and a European put option on a non-dividend paying stock S . You are given the following information:

- $r = 0.04$
- The current price of the call option $V_C(0)$ is by 0.15 greater than the current price of the put option $V_P(0)$.
- Both the put and the call expire in 4 years.
- The put and the call have the same strikes equal to 70.

Find the spot price $S(0)$ of the underlying asset.

- a. \$48.90
- b. \$59.80
- c. \$69.70
- d. \$79.60
- e. None of the above.

Problem 7.6. (10 points) A non-dividend-paying stock currently trades at \$50 per share. The continuously compounded risk-free interest rate is $r = 6\%$. A 1-year, \$50-strike European call on the stock currently sells for \$6.00, and the 1-year, \$50-strike European put currently sells for \$2.50. Which of the following strategies exploits an arbitrage opportunity, and what is the resulting risk-free profit in today's dollars?

- Buy the call, sell the put, short the stock, and lend; profit \$0.59
- Sell the call, buy the put, buy the stock, and borrow; profit \$0.59
- Sell the call, buy the put, buy the stock, and borrow; profit \$2.91
- Buy the call, sell the put, short the stock, and lend; profit \$2.91
- There is no arbitrage opportunity.

Problem 7.7. (5 points) A non-dividend-paying stock currently trades at \$40 per share. The continuously compounded risk-free interest rate is $r = 8\%$. You are given:

- A 1-year, \$40-strike European put on the stock costs \$2.72.
- A 6-month, \$38.43-strike European put on the stock costs \$1.90.

A chooser option, purchased today, allows its holder to decide in 6 months whether the option becomes a 1-year (from today), \$40-strike European call or a 1-year (from today), \$40-strike European put. Find the price of the chooser option today.

- \$4.62
- \$5.80
- \$7.20
- \$7.70
- None of the above.

Problem 7.8. (10 points) A non-dividend-paying stock currently trades at \$40 per share. The continuously compounded risk-free interest rate is $r = 8\%$. You are given the following market prices:

- A 1-year, \$40-strike European call on the stock costs \$5.80.
- A 6-month, \$38.43-strike European put on the stock costs \$1.90.
- A chooser option that allows its holder to decide in 6 months whether the option becomes a 1-year (from today), \$40-strike European call or a 1-year (from today), \$40-strike European put costs \$8.30.

Which of the following portfolios yields an arbitrage profit, and what is the resulting risk-free profit in today's dollars?

- Sell the chooser, buy the 1-year \$40-strike call, and buy the 6-month \$38.43-strike put; profit \$0.60
- Buy the chooser, sell the 1-year \$40-strike call, and sell the 6-month \$38.43-strike put; profit \$0.60
- Sell the chooser and buy the 1-year \$40-strike call; profit \$2.50
- Sell the chooser, buy the 1-year \$40-strike call, and buy the 6-month \$38.43-strike put; profit \$0.90
- There is no arbitrage opportunity.